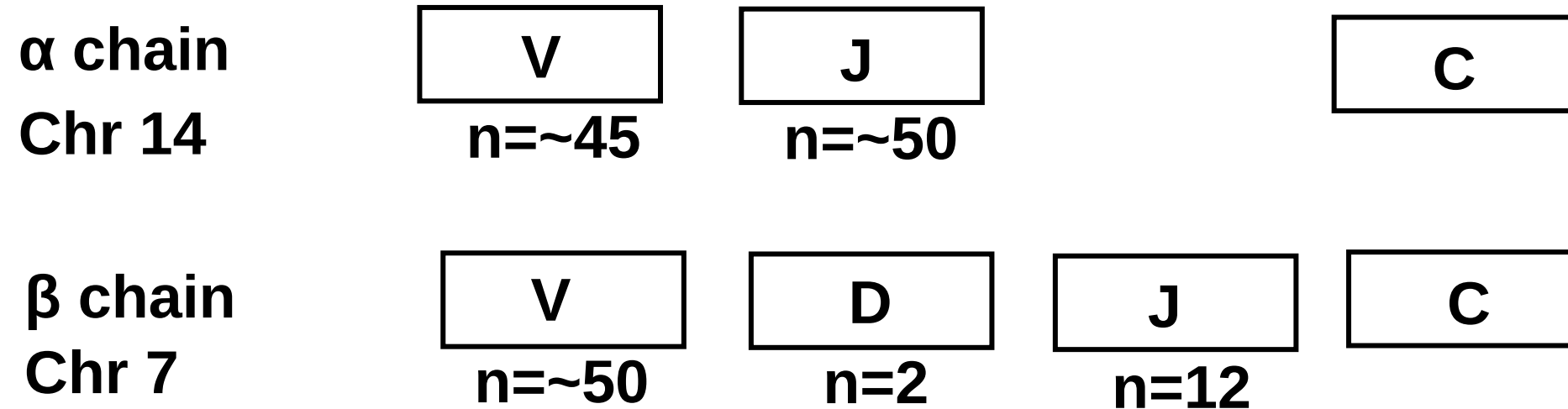


CDR1 and CDR2 encoded by germline in V



CDR3 by DJ and junctional for VDJ B and J + VJ junctional for a

**junctional diversity
(non-germline)**

largest contributor to diversity

P nucleotides (template after hairpin opening by Artemis)

N nucleotides (then nontemplate followed by DNA ligation) (20 by TdT) - more common

**combinatorial diversity
(germline)**

1 to 3 million upper limit)

10^{15} 10^{20} unique TCR combinations